

Swapnil Mehta

+91 8383045113 mehtaparth210@gmail.com linkedin.com/in/swapnil-mehta-sm

Professional Summary

AI/ML undergraduate with hands-on experience building Generative AI applications, Retrieval-Augmented Generation (RAG) systems, LLM-powered assistants, semantic search pipelines, and intelligent document processing solutions. Skilled in Python, SQL, Pandas, NumPy, LangChain, FastAPI, vector databases, and machine learning.

Education

Thapar Institute of Engineering & Technology (TIET), Patiala Aug 2023 - Present
B.Tech, Electronics and Communication Engineering CGPA: 8.02 / 10.0

Technical Skills

Programming Languages: Python, SQL, JavaScript, C++, C
AI/ML Technologies: Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Embeddings, Vector Databases (FAISS, ChromaDB), Machine Learning, Neural Networks, NLP, Model Evaluation, Hallucination Reduction, Feature Engineering, Data Analysis
Frameworks & Libraries: LangChain, FastAPI, Pandas, NumPy, Scikit-learn, OpenCV, YOLOv5, TFLite, React.js, Node.js, Express.js, Next.js
Tools & Platforms: PostgreSQL, MongoDB, MySQL, Git, GitHub, Jupyter Notebook, REST APIs, Postman, VS Code, Gemini API

Experience

- Technical Intern - AI & IT Systems | Food Safety and Standards Authority of India (FSSAI)** May 2026 - Jul 2026
- Developed a Retrieval-Augmented Generation (RAG) platform for food safety and regulatory documents using Python, embeddings, semantic retrieval, and LLM-powered question answering, improving information discovery across 5,000+ document pages.
 - Engineered intelligent document processing pipelines for 5,000+ pages of regulatory documents, implementing PDF extraction, semantic chunking, embedding generation, vector indexing, and contextual retrieval to improve document search efficiency by 40%.
 - Performed exploratory data analysis, data cleaning, and feature engineering on 100,000+ text records using Pandas and NumPy, improving retrieval relevance and model grounding quality by 20%.
 - Conducted prompt engineering, model evaluation, hallucination reduction testing, and response-quality benchmarking, improving answer relevance and context retention by an estimated 25%.
 - Authored technical documentation, API validation reports, and system testing artifacts while identifying and documenting 20+ critical defects across 3 government platforms.
- Computer Vision Research Intern | TIET - Experimental Learning Center.** May 2025 - Jul 2025
- Developed PCB defect detection pipeline using YOLOv5 and OpenCV achieving 89% mAP@0.5, 91% recall, and 83% precision through augmentation and hyperparameter tuning.
 - Reduced false positive rate by 30% via preprocessing pipelines and improved annotation efficiency by 40% across a 20-member research team.

Projects

- ResearchMind AI - Generative AI Research Assistant** | *Python, FastAPI, LangChain, Gemini, FAISS, PostgreSQL*
- Developed an end-to-end Generative AI platform enabling users to upload research papers and interact through citation-grounded natural language conversations.
 - Developed Retrieval-Augmented Generation (RAG) pipelines including PDF extraction, semantic chunking, embedding generation, vector indexing, and contextual retrieval using LangChain and FAISS.
 - Integrated Gemini APIs for intelligent document processing, research summarization, literature review generation, flashcard creation, and multi-document analysis.
- Dental Clinic SaaS - Smile Saviours** | *NestJS, React.js, Drizzle ORM, PostgreSQL, REST APIs*
- Built a multi-tenant clinic management platform supporting appointment scheduling, patient records, treatment tracking, billing workflows, and role-based access control.
 - Developed 15+ REST API endpoints using NestJS and PostgreSQL while implementing secure authentication and scalable backend architecture.
- Email Spam Classifier** | *Python, Scikit-learn, NLP, TF-IDF, Logistic Regression*
- Built NLP classification pipeline on 5,500+ labelled messages using TF-IDF feature extraction and Logistic Regression.
 - Achieved 98% classification accuracy and 0.96 F1-score through preprocessing, feature engineering, model validation, and performance evaluation.

Relevant Coursework

Data Structures & Algorithms | Machine Learning | Database Management Systems | Image Processing & Computer Vision | IoT-Based Systems

Leadership & Extracurricular

- Publicity Head - Girl Up UN Society, TIET** Aug 2025 - Aug 2027
- Directed a 70+ member team across 5+ campus initiatives and increased social media engagement by 25% through data-driven outreach strategies.
- Mudra Performing Arts Society, TIET** Dec 2023 - Dec 2024
- Anchored 4+ university events for audiences of 2,000+ and developed strong communication, collaboration, and public speaking skills.